Radiation Assurance For CubeSat Power System

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INTRODUCTION

Our mission is a 3 to 5 year Solar Synchronous Orbit (SSO), a near polar low Earth orbit (LEO) whose orbit allows our cubesat to maintain an approximate constant local solar time. Our planned orbit is 300-600 km with an inclination of ~98°.

To present the safest results, I will assume the worst case scenario. That is to say that I will set an explicit scope of 5 year mission at 600km at both solar max and min. Due to the high inclination of our orbit, we will repeatedly traverse the South Atlantic Anomaly (SAA), a region associated with elevated trapped particle fluxes.

This work focuses on radiation protection for specifically the battery management system (BMS) and the power distribution unit (PDU). Although general radiation protection plans could be adopted on any subsystem, this paper will focus specifically on the BMS and PDU. The criticality of these two parts will increase the need for risk reduction. This is because a failure of these systems will have cascading consequences and could result in mission failure.

Preliminary Radiation Information

Trapped Particles

Trapped protons and trapped electrons are high energy particles that are trapped by a planet's magnetic field, forming radiation belts. On Earth these are the Van Allen belt, they are primarily electrons which dominate the outer belt and protons which dominate the inner belt. Trapped electron radiation is modeled with AE-8 and trapped Proton radiation is modeled with AP-8. (Poivey & Day, 2002)

Solar Cycles (Max Solar Activity / Min Solar Activity)

The sun goes through ~11 year cycles in which solar activity increases and decreases. Sun spot activity is used to track solar cycle due to their being a greater number during increased solar activity. The peak of the sun's activity is solar maximum and when the sun is at its least active it is referred to as solar minimum. (Samwel et al., 2019) (Laurence et al., 2023)

The solar cycle contributes greatly to the composition of the orbital radiation environment. Our launch will likely be around a solar minimum. Trapped electrons and solar protons are lower, trapped protons and galactic cosmic rays (GCRs) are higher. SEUs attributed to trapped protons are higher, TID from protons will be more prevalent than TID from electrons (Samwel et al., 2019). These are generalizations and a more concrete launch window will allow for more accurate predictions.

Figure 1 is a chart tracking sunspot progression. Based on the data from NOAA the last solar maximum was around October of 2024, with the next solar minimum expected to take place between January 2029 and December 2032. *(Solar Cycle Progression | NOAA / NWS Space Weather Prediction Center, n.d.)*

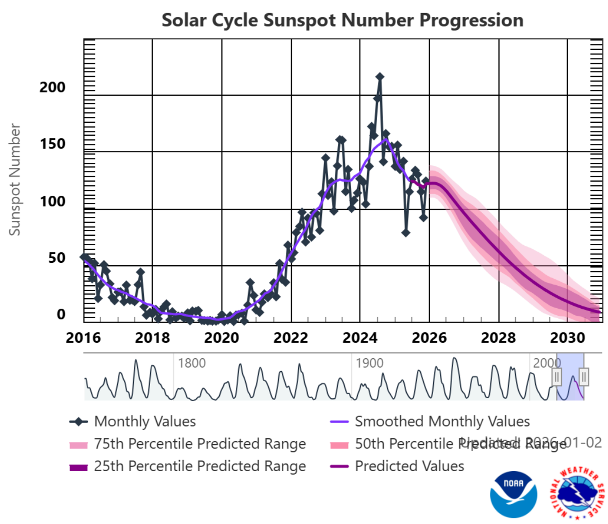

Figure 1 - Solar Cycle Sunspot Number Progression (Solar Cycle Progression | NOAA / NWS Space Weather Prediction Center, n.d.)

Solar Protons

Protons that are accelerated during solar events and are commonly associated with solar flares and coronal mass ejections (CMEs). These events contribute to both dose and single event effects(SEE). With our launch expected to take place around solar minimum we can expect these to be fewer in number (Poivey & Day, 2002).

Galactic Cosmic Rays (GCRs)

Energetic protons and heavy ions that have unknown origins but are believed to come from celestial events such as supernovas. They cause secondary radiation called Bremsstrahlung or X-rays from contact with the electric field of an atomic nucleus. The secondary radiation is often difficult to block which will lead to SEE (Rahman et al., 2017).

Van Allen Belts

The Van Allen Belts are two toroidal regions of trapped particles surrounding earth. They are classified as the inner and outer belts. The outer belt consists mainly of electrons with $E < 7$ MeV and the inner belt consists mainly of protons with $E < 500$ MeV (Samwel et al., 2019). The outerbelt traps most celestial radiation, making it the most dangerous. Our orbit will have us mainly traverse within the inner belt although we will have to pass through the South Atlantic Anomaly (SAA) and polar horns. Passing through these regions will increase exposure to trapped electrons, solar protons, and low energy GCRs. (NPRP, 1996)

Polar Horns

“The extensions of the outer radiation belts to low altitude”(Samwel et al., 2019). Due to the shape of the radiation belts polar orbiting satellites will pass through areas of the outer belt increasing radiation exposure and risk factors.

South Atlantic Anomaly (SAA)

An area above the atlantic where the outer belt is closer to the Earth which is caused by the orbit and tilt of the Earth's geomagnetic axis (Samwel et al., 2019). This is a historic cause of SEE problems in components. (NPRP, 1996)

Total Ionising Dose (TID)

Is a cumulative metric describing the total ionising radiation energy absorbed by a material and is most commonly measured in rads. It is the metric used to describe the slow damage caused by exposure to radiation. It can be contributed to by secondary Bremsstrahlung photons and solar flare protons although in LEO the primary concern is trapped electrons and protons in the inner belts (Rahman et al., 2017).

Single Event Effect (SEE)

Is a metric used to describe the damage done when a single particle such as a heavy ion or a high energy proton interacts with a semiconductor. Additional shielding thickness does not do much to mitigate damage after the initial strike due to the high energy of the particles. SEE is characterized using Linear Energy Transfer or $MeV \cdot cm^2/mg$ (Poivey & Day, 2002)(Rahman et al., 2017).

SEE has several subtypes, single event upset (SEU), single event latch up (SEL), and single event burnout (SEB). SEU is a transient and non-destructive radiation induced “soft error” such as a bit flip. SEL is when a parasitic conductive path forces a gate high until power is removed, this is extremely common in CMOS. SEB is when the energised particle causes high current leading to component burnout (Poivey & Day, 2002)(Rahman et al., 2017)(NPRP, 1996).

Displacement Damage Dose (DDD)

Is cumulative non-ionizing damage which is caused by the collision between an incoming particle and a lattice atom, which results in the atom being displaced. For this to happen the incoming particle must have enough kinetic energy to break the chemical bonds holding the atom in place and move it far enough so that it cannot return. In LEO this is primarily caused by trapped and solar protons. DDD is measured using non-ionising energy loss (NIEL) which is $MeV\ *\ cm^2/g$ or $MeV\ *\ g(Si)^{-1}$ (Samwel et al., 2019).

Radiation Analysis

We will need to conduct our own radiation predictions, however due to the time constraints of this assignment and lack of a fixed launch window I was not able to incorporate them into this report. As a result I will have to go off of data from previous missions. This can be used to provide a base line assumption to dictate design policy.

TID is the result of prolonged exposure to ionizing radiation. Most of the particles that contribute to TID can be reduced by a buffer of mass between the electronics and space. Internal radiation depends on external radiation and the amount of shielding we can provide. A CubeSat with our orbit will commonly have a 0.204 cm outer wall thickness, with 0.15 cm being an absolute minimum (Laurence et al., 2023) *(Samwel et al., 2019)*. Figure 2 demonstrates the absorbed radiation in a sphere of aluminium with a similar mission profile to our own. Notice how increased shielding is more effective at blocking electrons than protons. (Rahman et al., 2017). Additional shielding can be placed around components which are of greater risk or importance. Vaults can be constructed as well although for most smallsats are not commonly used.

These findings are corroborated by the findings of other research papers. Figure 3 demonstrates the “TID, DDD, and SEU during maximum and minimum solar activity, using no shielding, and 1.5 mm aluminum shielding”(Samwel et al., 2019). While this study concluded that for their mission a combination of circuit design, COTS components, and 1.5 mm Al shielding was optimal, based on how these numbers compare to part tolerances this is due to their mission profile being only 3 years. Mission time over 3 years requires a more conservative design philosophy.

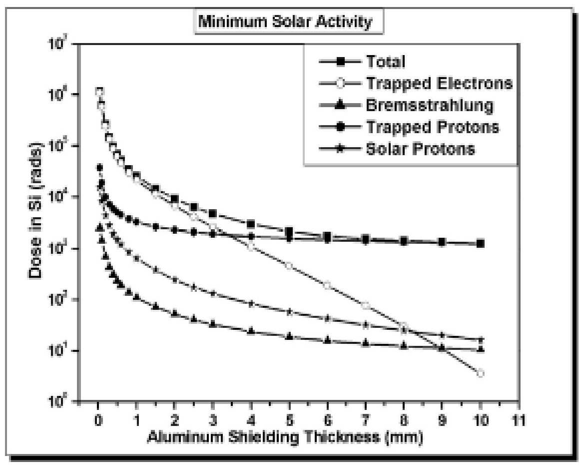

Figure 2 - Dose depth curves in Si as a target material, for a 3 year mission length, during minimum solar activity (Rahman et al., 2017).

Effect	Solar activity	No shielding	1.5 mm
TID (rads)	Maximum	3.73×10^6	3.31×10^3
	Minimum	2.63×10^6	2.29×10^3
DDD (MeV g(Si)$^{-1}$)	Maximum	1.75×10^{11}	6.64×10^7
	Minimum	1.24×10^{11}	5.93×10^7
SEU (Bit^{-1})	Maximum	10^{-2}–10^2	10^{-6}–10^{-2}
	Minimum	10^{-2}–10^2	10^{-7}–10^{-2}

Figure 3 - TID, DDD, and SEU during maximum and minimum solar activity, using no shielding, and 1.5 mm aluminum shielding (Samwel et al., 2019).

While increased shielding will result in increased TID protection, DDD and SEE are not the same. SEE mitigation is primarily circuit level. Design choices can limit logic risks and failure rates. Some design implementations used to mitigate SEE are by preferring analog design, decreasing activity time, implementing voting, watchdog timers, redundancy, and lockstep into our circuit design (Rahman et al., 2017)(Poivey & Day, 2002).

Voting and redundancy are similar concepts in which additional devices are used to perform the same task to ensure a valid state. Voting refers to the “Use three or more devices to perform the same function. If one device disagrees with the rest, use the remaining devices to determine the next state. ” (Rahman et al., 2017). Redundancy is the same, however it compares only two devices. Watchdogs timers are devices used to monitor other devices. They check the device for corruption, in the event there is corruption the corrupted device is reset. Lockstep refers to two devices with simultaneous clock and inputs, if there is a discrepancy between the two devices they are reset. Finally repetition is a technique which refers to a device performing repeated tasks. The results of those tasks are compared against each other and must achieve the same sequential result for an action to be valid.(Rahman et al., 2017)(Poivey & Day, 2002)

System Overview

The BMS is responsible for ensuring safe charge and discharge of the batteries and ensuring their protection and longevity. Its key functions are as follows

1. Cell balancing
2. Over-current (OC) protection
3. Over-voltage (OV) protection
4. Under-voltage (UV) protection
5. Short-circuit (SC) protection
6. Temperature protection

Similarly, the PDU must distribute power safely and reliably to the designated output rails. It typically implements OC/OV/UV protection and load switching/current limiting per rail. Figure 4 details a generalized block diagram of a BMS/PDU circuit. Parts of the solar panel subsystem will likely end up on the final PCB, however for the focus of this research have been omitted from this analysis.

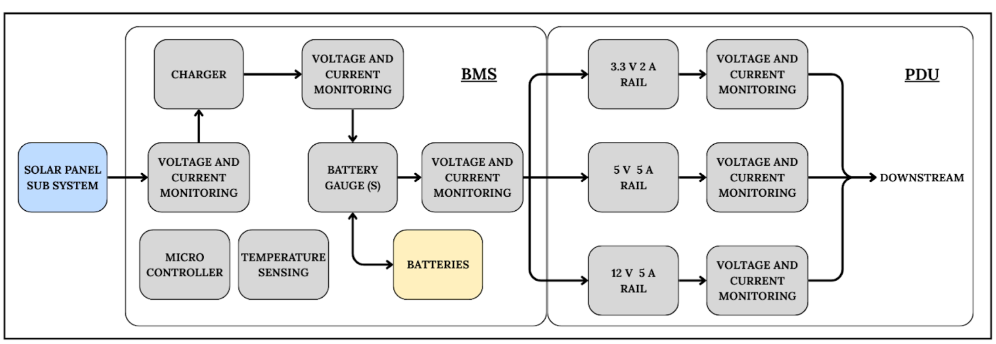

figure 4 - Generalised BMS/PDU block diagram

Threat analysis of the BMS/PDU

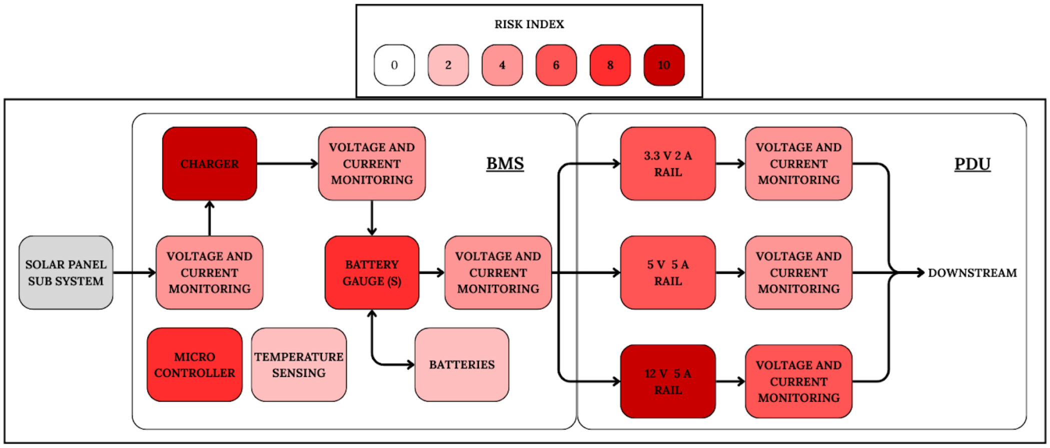

figure 5 - Generalised BMS/PDU with Risk Assessment per Component

The risk sensitivity of each component is highlighted in *figure 5* With 0 being minimum danger and 10 being maximum danger. This assessment was based on component anatomy, risk exposure, as well as scale of consequence in failure. Components that required complex digital implementation, elevated exposure, or contained high power gates resulted in an elevated score. Components with simple analog implementation, low exposure, or contained low power gates resulted in a lower score. The following are core takeaways.

1. Analog solutions should be incorporated where possible to diminish risk of bit errors. In our previous research we concluded that while preferring analog design decreases radiation risk, its implementation is more complex, which could expose us to elevated risk of human error. When digital design must be used, assurance policies need to be put in place to maintain the integrity of the code.

2. Increased periods of device activity leave the components more vulnerable to time independent damage such as latch-up or bit disturbances.

3. A latch up failure in components controlling power flow causes catastrophic downstream consequences. This is most prevalent for the charger and the 12v rail but is true generally. (Demirköz, Melahat Bilge et al., 2025)

The charger and microcontroller are instrumental in handling and enforcing the duty cycle. A fault within the components relating to the duty cycle would be catastrophic due to the large amount of power through the component and the sensitivity of downstream components. These parts are too complex to be made analog. As a result the component will be in frequent use and its prolonged on time will result in increased risk from both time TID and SEE.

Figure 6 details conservative estimates of radiation tolerances based on component grade. The data comes from (NPRP, 1996) (Rahman et al., 2017). It is important to note that COTS are not likely to be radiation tested and that these numbers are generalisations. While Rad-tollerant and Rad-hardened components should be tested and list their radiation tolerances in their data sheet, research has shown that these numbers are not to be taken at face value (Demirköz, Melahat Bilge et al., 2025).

	TID (krad(Si))	SEE
COTS	≥ 2 - 10 krad	5 MeV·cm²/mg
Rad-tolerant	≥ 20 - 50 krad	20 MeV·cm²/mg
Rad-hard	≥ 200 - 1000 krad	37 MeV·cm²/mg

figure 6 - Conservative Radiation Tolerances as per Component Grade (NPRP, 1996) (Rahman et al., 2017)

Conclusion

The results of my assessment of radiation assurance for our Cube Sat's power system are as follows. Although COTS may be acceptable for a 3 year mission provided adequate design choice, this comes with the risk of untested components. In order to maximize mission safety, rad tolerant component selection is advised alongside preferring analog design, decreasing activity time, implementing voting, watchdog timers, redundancy, and lockstep into our circuit design.

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